ECON100B - SECTION 18

PSET Practice and Intro to Monetary Policy and Phillips Curve

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ANNOUNCEMENTS/AGENDA

Announcements

- PSET 3 released: Due on Friday, 4/17

Agenda

- Problem Set 3
- Monetary Policy and the Phillips Curve

Question of the Day

Using the IS Model, we are told that our government spending is 1% higher than expected, with all other factors of our national income identity equation being at expected levels. We are told that our MPK is 4% and our real interest rate is 3%, with a sensitivity parameter $\bar{b}$ of 0.5. We are also told that we have a multiplier effect/marginal propensity to consume of 0.25.

What is our calculated short run output given this information?

Question of the Day Answer

Our increase in government spending tells us that our aggregate demand is equal to 1%. Our MPK – real interest rate difference along with the sensitivity parameter means that the real interest rate increases our short run output by 0.5%. Using our multiplier, we find a total **short run output of 2%.**

PSET 3, Q4

4. [12 points total] The IS curve, interest rates, and macroeconomic shocks.

Recall how the IS curve is a downward-sloping line in $\left(\tilde{Y}, R\right)$-space

$$\tilde{Y} = \bar{a} - \bar{b}\,(R - \bar{r}) \qquad (4.1)$$

(a) [2 points] Graph the IS curve on the axes below, being careful to position it correctly so that you are *depicting no temporary shocks* to aggregate demand.

PSET 3, Q4

Suppose that $\bar{b} = 0.4$ and $\bar{r} = 0.03$ so that

$$\tilde{Y} = \bar{a} - 0.4\,(R - 0.03) \qquad (4.2)$$

In each of the following scenarios, discuss what happens to the IS curve, if anything, and to short-run output $\tilde{Y}$, if anything. Begin each scenario by assuming that there is no temporary shock to aggregate demand $\bar{a} = 0$ and that monetary policy is neutral, $R = \bar{r}$. Both may change.

(b) [2 points] Suppose consumers become spooked by falling stock prices and temporarily reduce consumption spending by 0.1 percentage point of potential GDP.

(c) [2 points] Suppose the Fed decides it must fight inflation (not shown here), and it raises the real interest rate to $R' = 0.04$.

PSET 3, Q4

(d) [2 points] Suppose Congress (the government) votes to go to war and passes a bill to raise government spending on military goods and services by $0.3 trillion or roughly 1 percent of GDP, and the President signs the bill.

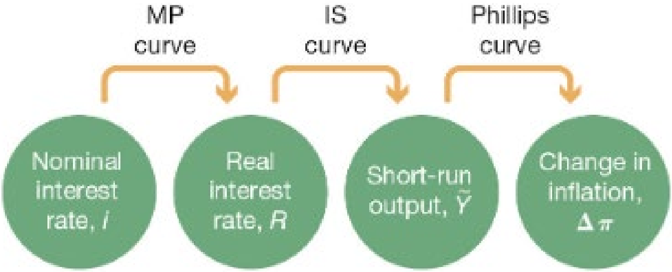
(e) [2 points] Suppose foreign economies temporarily increase their demand for U.S. exports by 0.3% of potential GDP.

(f) [2 points] Suppose the marginal product of capital $\bar{r}$ falls from $\bar{r} = 0.03$ to $\bar{r}' = 0.02$ but real interest rates remain at $R = 0.03$.

Monetary Policy and the Phillips Curve

The Central Bank and Monetary Policy

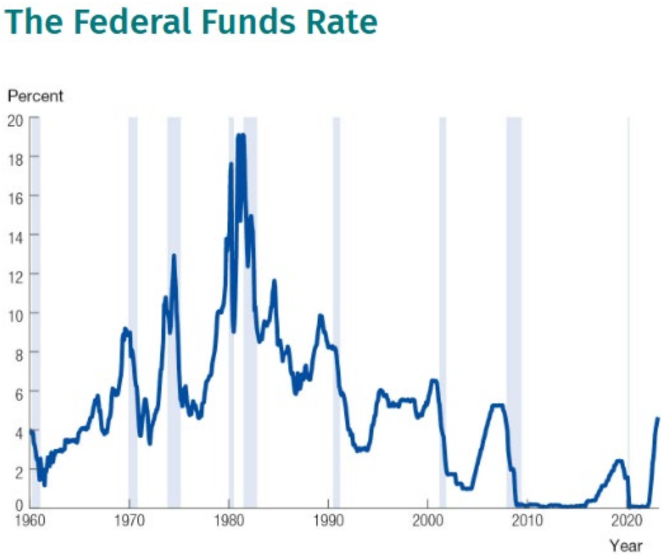
- The main policy tool used by the U.S. Federal Reserve is the federal funds rate, which is the interest rate paid from one bank to another for overnight loans



How does the Federal Funds Rate Work?

- Banks and financial institutions routinely lend to and borrow from one another daily for several reasons
- The Central Bank is able to set this rate by stating that they are willing to borrow or lend any amount at a specific rate. In the U.S., this is the federal funds rate
- Every bank will charge this same rate on overnight loans, otherwise banks would just borrow at the lower rate from the central bank
- The Central Bank’s willingness to borrow at this same rate helps to disincentivize any bank from charging a lower overnight loan rate

The Federal Funds Rate over Time



Recalling the Fisher Equation

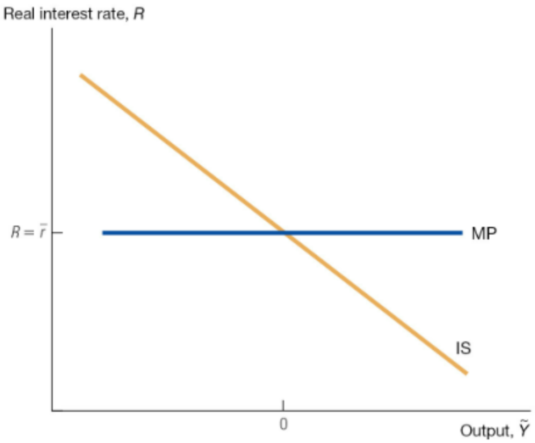
$$i_t = R_t + \pi_t.$$

$$R_t = i_t - \pi_t.$$

- Changes in the nominal interest rate will lead to changes in the real interest rate as long as they are no offset by corresponding changes in inflation
- The sticky inflation assumption states that inflation takes time to change and will not do so immediately
- Practically, what this means is that central banks set the real interest rate in the short run

The IS/MP Diagram

The MP Curve in the IS/MP Diagram



The Phillips curve, revisited

Economy-wide inflation will be equal to what firm owners across the economy expect inflation will be, plus/minus the effect of any booms or busts

$$\pi_t = \underbrace{\pi_t^e}_{\text{expected inflation}} + \underbrace{\bar{\nu}\tilde{Y}_t}_{\text{demand conditions}} .$$

with the assumption:

$$\pi_t^e = \pi_{t-1}.$$

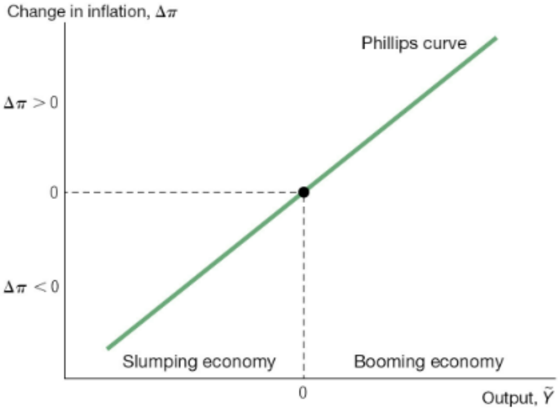
So we can rewrite our Phillips Curve

$$\pi_t = \pi_{t-1} + \bar{\nu}\tilde{Y}_t.$$

$$\Delta\pi_t = \bar{\nu}\tilde{Y}_t.$$

The Phillips curve

The Phillips Curve



Unexpected Shocks

- Our model of inflation currently considers previous inflation/expected inflation and the state of the economy
- But what about sudden price shocks? For example, the current conflict in the Middle East?
- We introduce a shock term to inflation as well:

$$\pi_t = \pi_{t-1} + \bar{\nu}\tilde{Y}_t + \bar{o}.$$

$$\Delta\pi_t = \bar{\nu}\tilde{Y}_t + \bar{o}.$$

- This is the final form of the Phillips curve which we'll use

Types of Inflation

- Cost-push inflation: Inflation that is caused because the costs of inputs push the costs of goods and services up
- Demand-pull inflation: Changes in the aggregate demand in the economy affect the inflation rate

Phillips Curve Practice Problems

1. If we observe a shock in oil prices, what happens to the movement of our Phillips Curve compared to baseline?
2. Under the Phillips Curve framework, what component of the Phillips Curve is considered cost-push inflation, and which component is considered demand-pull inflation?
3. We are told that inflation in the previous period was 4%, and our short run output is expected to be -2% with a sensitivity parameter of 0.5. We are also hit with a sudden supply chain shock that has increased a 1.5% shock to inflation. What is our final inflation rate in this current period? How would we show this on a graph with the Phillips Curve?

Phillips Curve Practice Solutions

1. Our Phillips Curve pushes left and out: we expect to see higher levels of inflation for any short-run output levels because of the oil price shock.
2. Shocks or “o” are considered cost-push inflation and short-run output effects are considered demand-pull inflation
3. We’d expect a 0.5% inflation rate. Our curve pushes left-ward and outward, but we also move down the new curve.

Modeling the 1970s Inflation Crisis using the Complete Short-Run Model